Configuração 12: $M = 256$, $C = 8$, $K = 12$, IoT = 20 dB, 16-QAM
Perda de SNR do BCD-MMSE (1 iteração) em SER = 10^{-2}, em dB, em relação à referência ideal (float64); × = não chega a 10^{-2} nas SNRs simuladas
Fundo laranja no rótulo: o modo com o melhor resultado no par, comparando do formato com menos bits para o de mais, sem contar as colunas em que todos os modos ficam abaixo de 0,05 dB
Embaixo da perda: maior ocupação da XCU50 entre LUT, DSP, FF e BRAM (síntese HLS, sem diretivas), no mesmo formato e modo

Perda de SNR (dB)

< 0,1 dB | 0,1 a 0,5 dB | 0,5 a 1 dB | 1 a 2 dB | 2 a 3 dB | 3 a 5 dB | 5 a 10 dB | > 10 dB | × não chega a 10^{-2} | rótulo do melhor modo

TDLA30-5

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	1.53 1.8%	0.00 2.1%	0.00 3.1%	0.00 3.6%
	AP_SAT_ZERO	1.53 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.53 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.53 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	1.53 1.8%	0.00 2.2%	0.00 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	2.29 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.29 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	2.29 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	2.29 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	2.29 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	0.06 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.06 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.06 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.06 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.06 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	0.06 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.06 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.06 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.06 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.06 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	0.06 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.06 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.06 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.06 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.06 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	0.06 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.06 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.06 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.06 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.06 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	0.06 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.06 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.06 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.06 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.06 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%

TDLA30-75

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	1.61 1.8%	0.00 2.1%	0.00 3.1%	0.00 3.6%
	AP_SAT_ZERO	1.61 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.61 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.61 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	1.61 1.8%	0.00 2.2%	0.00 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	2.23 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.23 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	2.23 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	2.23 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	2.23 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	0.06 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.06 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.06 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.06 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.06 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	0.06 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.06 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.06 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.06 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.06 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	0.06 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.06 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.06 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.06 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.06 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	0.06 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.06 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.06 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.06 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.06 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	0.06 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.06 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.06 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.06 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.06 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%

TDLA30-300

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	1.42 1.8%	0.00 2.1%	0.00 3.1%	0.00 3.6%
	AP_SAT_ZERO	1.42 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.42 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.42 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	1.42 1.8%	0.00 2.2%	0.00 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	2.26 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.26 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	2.26 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	2.26 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	2.26 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	0.07 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.07 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.07 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.07 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.07 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	0.08 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.08 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.08 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.08 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.08 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	0.07 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.07 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.07 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.07 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.07 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	0.07 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.07 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.07 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.07 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.07 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	0.07 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.07 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.07 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.07 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.07 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%

TDLC300-100

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	1.25 1.8%	0.01 2.1%	0.01 3.1%	0.00 3.6%
	AP_SAT_ZERO	1.25 1.8%	0.01 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	1.25 1.9%	0.01 2.2%	0.01 3.7%	0.00 4.2%
	AP_WRAP	1.25 1.7%	0.01 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	1.25 1.8%	0.01 2.2%	0.01 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	2.19 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.19 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	2.19 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	2.19 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	2.19 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	0.08 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.08 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.08 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.08 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.08 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	0.07 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.07 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.07 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.07 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.07 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	0.07 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.07 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.07 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.07 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.07 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	0.07 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.07 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.07 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.07 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.07 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	0.08 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.08 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.08 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.08 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.08 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%

TDLB100-400

Quantização	Estouro	<22,10>	<26,10>	<32,12>	<36,12>
AP_TRN	AP_SAT	0.24 1.7%	0.00 2.0%	0.00 3.3%	0.00 3.7%
	AP_SAT_ZERO	0.24 1.6%	0.00 2.2%	0.00 3.9%	0.00 4.2%
	AP_SAT_SYM	0.24 1.8%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP	0.24 1.6%	0.00 2.1%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.34 1.7%	0.18 2.2%	0.00 3.9%	0.00 4.4%
AP_TRN_ZERO	AP_SAT	2.15 1.9%	0.01 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	2.15 1.7%	0.01 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	2.15 2.1%	0.01 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	2.15 1.4%	0.01 1.9%	0.00 3.4%	0.00 3.9%
	AP_WRAP_SM	2.15 1.9%	0.19 2.3%	0.00 3.9%	0.00 4.4%
AP_RND	AP_SAT	0.05 1.8%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	0.05 1.7%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	0.05 2.0%	0.00 2.4%	0.00 3.9%	0.00 4.4%
	AP_WRAP	0.05 1.6%	0.00 2.1%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.22 1.9%	0.14 2.2%	0.00 3.9%	0.00 4.4%
AP_RND_ZERO	AP_SAT	0.05 1.9%	0.00 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	0.04 1.7%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	0.05 2.1%	0.00 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	0.05 1.4%	0.00 1.9%	0.00 3.4%	0.00 3.9%
	AP_WRAP_SM	0.18 1.9%	0.14 2.3%	0.00 3.9%	0.00 4.4%
AP_RND_MIN_INF	AP_SAT	0.05 1.8%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	0.05 1.7%	0.00 2.2%	0.00 3.9%	0.00 4.2%
	AP_SAT_SYM	0.05 2.0%	0.00 2.4%	0.00 3.9%	0.00 4.3%
	AP_WRAP	0.05 1.6%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP_SM	0.18 1.9%	0.14 2.2%	0.00 3.9%	0.00 4.4%
AP_RND_INF	AP_SAT	0.05 1.9%	0.00 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	0.05 1.7%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	0.05 2.1%	0.00 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	0.05 1.4%	0.00 1.9%	0.00 3.4%	0.00 3.9%
	AP_WRAP_SM	0.18 1.9%	0.14 2.3%	0.00 3.9%	0.00 4.4%
AP_RND_CONV	AP_SAT	0.05 1.9%	0.00 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	0.05 1.7%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	0.05 2.1%	0.00 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	0.05 1.6%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP_SM	0.18 1.9%	0.14 2.3%	0.00 3.9%	0.00 4.4%